Sreeyutha Ratala and Erica Wang

Network Properties in the Epstein Files

Motivation

The social networks we studied in class have generally been innocuous, such as the authorship network of Caltech professors or the connections between two unrelated YouTube videos. For such networks, our primary motivation was to identify common network properties (such as small diameter, heavy-tailed distributions, strongly connected components, and potentially high levels of clustering) rather than to perform an in-depth analysis of which specific nodes play interesting or noteworthy roles or of how different components differ in structure and in the properties they identify. For example, one cluster or component may be composed primarily of nodes sharing one common attribute (e.g., institutional affiliation), while another may display a different attribute.

Interest in such further analysis may be piqued by additional contextual considerations, such as broader societal, moral, and political factors correlated with a network's structure. One real-world example of a social network is the email communication network formed by Jeffrey Epstein (jeevacation@gmail.com), spanning from the early 2000s to the late 2010s, as documented in the publicly accessible, redacted Epstein Files posted by the DOJ (Department of Justice).

What we call the Epstein File Network provides great value as a window into the structure of connectivity among individuals occupying positions of significant political, economic, academic, and social influence. Such connections are often opaque and difficult to observe, so analyzing them could reveal valuable insights into the structure of influence in the US political, economic, and social spheres. We offer a case study that may interest much of the public regarding this topic.

Furthermore, the Epstein Files are widely popular, and many excerpts are posted on major social media platforms, yet the information is limited and unbalanced. There is yet (to our knowledge) a more comprehensive study. Through this case study, we offer the public a more complete understanding of the Epstein Files and the network of people surrounding Jeffrey Epstein. This increases public information literacy and encourages improved critical evaluation of publicized events and the power structures behind them. We may also gain insights into why so many individuals implicated in the files avoid scrutiny by the authorities despite their obvious involvement in these highest of crimes. Although we can vaguely answer questions about why criminal prosecution is difficult, as the law requires proof beyond a reasonable doubt, combining this with our results can offer further perspective[1].

Goal

We can interpret each person mentioned in the Epstein Files as a node. We can build edges between two node individuals if they have been mentioned in the same email or document. Each node is connected to Jeffrey Epstein, as he is the center of the email communications. Thus, we will form an undirected graph.

Our goal is defined as follows:

1) Investigate different notions of centrality, such as degree, betweenness, and closeness centrality. We may also define our own versions of centrality based on how we define a node's importance. From this, we can identify critical nodes to investigate. These nodes may play critical roles in the network.

---

[1] https://www.law.cornell.edu/wex/beyond_a_reasonable_doubt

We suspect that Jeffrey Epstein was not the "main man in control," but rather played a role more akin to an administrative assistant in charge of communications and logistics. Investigations have identified numerous alleged co-conspirators and facilitators, suggesting Epstein operated within a larger network rather than in isolation.[2] It would be interesting to see whether we can identify notions of centrality that help us uncover nodes (i.e., people) that are less connected but still closely related to other nodes, thereby exerting influence over much of the network.

2) Identify different clusters, then, based on research into the nodes within each cluster, report how the individual nodes in different clusters are similar or different. Based on how different clusters are connected and interact with each other, form subjective notes on what roles different clusters may have served, both with regard to each other and the central node, Jeffrey Epstein.
3) Note if some nodes display more heavy-tailed properties than others, and if they are in specific communities. Are these nodes more connected? How are sub-hubs formed?
4) If we want to investigate the role of Jeffrey Epstein himself, we must note that this network is centered on him, so we will need to expand the scope. To do so, we must web-crawl a large graph that is not limited only to the Epstein files. We consider this task to be a bit more difficult, and we will revisit it in CS145. Once we do, we can analyze the properties that the Jeffrey Epstein node may fulfill.
5) And finally, due to the redacted nature of the files, it may be of interest to us to identify the identities of those redacted nodes based on insights from a more comprehensive view of his connections that our network graph may reveal.

Details

Information about the emails sent and received by Jeffrey Epstein is publicly available via the DOJ (Department of Justice) files[3]. It is heavily redacted, so our network graph will not be complete.

We will populate our Epstein File Network by performing a web crawl (coded by us) on the PDF files downloaded from the DOJ. Our crawler will search for names of individuals and link them together if they appear on the same PDF. Since the Epstein files also include images, we may also use a large language model to process the images into individual names.

For goal number 4, we may outsource files or websites on the internet for details that may build a larger network that outscopes the focus on Jeffrey Epstein. For this, we may code up a sort of website crawler similar to the ones we have studied in class.

End Product and Results

Our end product will be a paper or report of the different properties we notice and the potential conclusions we may draw. We intend to make this report publicly available for further work.

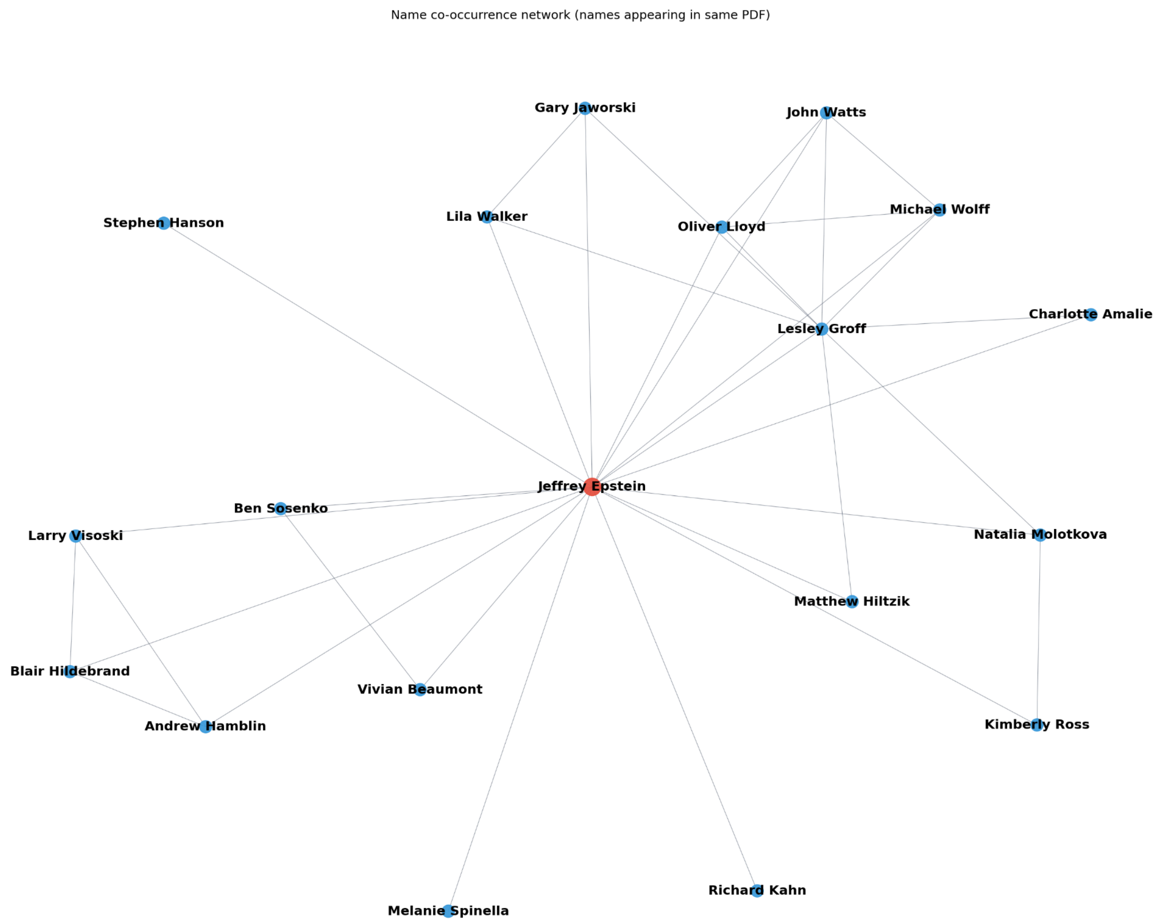
It is important to note that this is a very subjective study and is intended only as a thought piece for individual opinion, not to form any definitive conclusions or inform factual decisions.

Proof of Concept

[2] https://people.com/fbi-identified-10-alleged-co-conspirators-jeffrey-epstein-11876214

[3] https://www.justice.gov/epstein

For our proof of concept, we downloaded ~20 files from the Epstein files. We coded a pdf crawler, which detects names in an email chain. For each PDF (email chain), for each name detected, we create an edge between the two names, which we set as nodes in our graph. For each name, we also add an edge to Jeffrey Epstein. Image below.



We see the beginnings of clusters as well as likely-to-be-central nodes, so we believe that the network properties that we intend to investigate are well-founded. We furthermore look into the far left cluster, including Larry Visoski, Blair Hildebrand, and Andrew Hamblin. Larry Visoski, from a Google search, is Jeffrey Epstein's long-term pilot. Blair Hildebrand is a VP in Operations/Sales at New United Goderich, and Andrew Hamblin, who appears to be Director of Sales and Marketing at New United Goderich. New United Goderich is an aircraft maintenance company. Thus, this particular cluster seems to be focused on the task of fixing up Epstein's private plane. This serves as an example of the beginnings of the analysis we could do with a full study and the full graph.

Proof of concept code link: https://github.com/SreeyuR/cs144-proposal

Network-Aware Discovery in Second-Hand Fashion Ecosystems

Motivation

The second-hand fashion ecosystem has grown rapidly in recent years, with industry reports projecting the global resale market to expand at double-digit annual growth rates.[4] Despite this rapid growth, digital discovery in the resale ecosystem remains fragmented across independent sellers, local markets, and online communities, making it difficult to efficiently search for buyers seeking specific styles or unique items. While various tools exist to help individuals track finds or browse thrift store listings, many approaches treat sellers and buyers as isolated entities rather than recognizing the interconnected communities that shape how second-hand fashion is discovered and shared.[5] Existing platforms, such as Depop, Poshmark, and Vinted, emphasize individual listings and keyword search but do not model the broader network structure of fashion communities. Also, prior work highlights that fashion recommender systems aim to help buyers navigate large fashion catalogs, yet many existing approaches rely heavily on item features rather than modeling broader social or structural networks.[6] This gap presents an opportunity to rethink second-hand fashion discovery as a network problem rather than just as a search task.

We can model second-hand fashion discovery as a network problem because fashion culture is widely understood as a socially driven phenomenon, where trends and preferences spread through interconnected communities and word-of-mouth interactions rather than solely isolated individual behavior.[7] This makes it difficult for individual buyers to find “hidden gem” sellers and the products they offer, without prior knowledge of trends or locations, leaving many small vendors hidden within otherwise vibrant local ecosystems. In this regard, we define a “hidden gem” as a product that delivers exceptionally high satisfaction for its price. Many thrift stores and flea market vendors possess hidden gems within their unique inventory, yet lack the resources to maintain full online storefronts, leaving them disconnected from digital marketplaces.

Our application-driven project, ThriftNet, aims to address this gap by modeling the thrift ecosystem as an interconnected graph that combines structural relationships between sellers with social discovery patterns among buyers. By leveraging publicly available data sources, community contributions, and data from buyer participation in our application, the project explores how network-based modeling can help people discover amazing finds more effectively by reducing the effort required to find the items they want. Ultimately, this work investigates whether incorporating network structure can improve accessibility and discovery in decentralized resale marketplaces.

Goal

The goal of this project is to design and evaluate a social network-aware recommender system for the second-hand fashion ecosystem and determine whether it improves upon existing methods. While the broader vision applies to resale communities at large, we will construct a case study centered on Pasadena-area second-hand sellers, using

[4] https://www.futuremarketinsights.com/reports/second-hand-fashion-market
[5] https://www.mdpi.com/2071-1050/10/5/1474
[6] https://arxiv.org/pdf/2202.02757
[7] https://www.mdpi.com/2071-1050/10/5/1474

second-hand seller data from social media platforms[8], publicly available thrift store data[9], and social communities on online platforms (friends on our ThriftNet app which can also be curated using social media networks we have access to, such as Instagram).

We model this system using two interconnected networks, one social network with sellers as nodes and one seller network with buyers as nodes, connected through properties like geographic proximity, stylistic similarity, and social relationships. Real-world social networks often exhibit the small-world property, where short paths exist between most pairs of nodes. Leveraging this property allows buyers to discover new sellers through short paths in the social network (e.g., friends or friends-of-friends, see Fig. 4), reducing the search effort required to uncover hidden gem sellers. We exploit network structure using centrality measures and community detection to identify influential sellers and stylistic communities. In particular, clustering algorithms (such as spectral clustering or stochastic block model-based approaches) allow us to identify groups of sellers with similar aesthetic styles, revealing stylistic submarkets within the thrift ecosystem. We also study information diffusion by examining how buyers' experiences discovering "hidden gems" influence information propagation within friendship circles and local communities, similar to diffusion processes studied in social networks where information spreads along edges through repeated interactions between connected nodes.

The thrift ecosystem naturally forms a layered network: stores influence one another's inventory based on emerging fashion trends and successful sales (for example, one store's idea may inspire others to adapt or replicate it at a lower price), buyers move through stylistic clusters aligned with their preferences, and fashion trends diffuse through interconnected communities. By incorporating both marketplace structure (inherent to a store through its styles, inventory, and aesthetic) and social discovery (influenced by friends and extended social networks), ThriftNet evaluates the conditions under which network-aware recommendations outperform traditional keyword-based search. Here, keyword-based search refers to discovery methods that rely primarily on matching query terms to listings or store descriptions, rather than explicitly modeling the relationships within the second-hand fashion ecosystem among sellers, buyers, and between. In contrast, our approach incorporates factors such as stylistic similarity between sellers and social connections between users. We evaluate performance in terms of hidden-gem discovery rate, recommendation novelty, and effectiveness across diverse buyer profiles, including trend-unfamiliar buyers, buyers with highly individualistic styles, and socially trend-focused buyers.

Details

We model this problem as a heterogeneous network of buyers and sellers, connected through both marketplace structural similarity and social interactions. Seller nodes (stores/vendors) are connected through edges, based on whether sellers are similar enough under the following conditions: they either share at least T number of tags, are in the same area, or both have the same style type (i.e., curated/vintage/designer, etc.). Hence, the seller network is constructed based on geographic proximity and stylistic similarity, which is also influenced by trend-following behavior. This forms a network that captures how sellers relate within the second-hand fashion landscape.[10] We will also analyze structural properties of the resulting graph, such as the clustering coefficient, which measures the tendency for sellers within a stylistic niche or geographic area to form tightly connected groups. Note that this will

---

[8] https://www.instagram.com/p/DSxsWYAD04A/

[9] https://www.thriftmaps.app/

[10] https://pmc.ncbi.nlm.nih.gov/articles/PMC12367692/, https://www.researchgate.net/publication/353485380_Fashion_Recommendation_Systems_Models_and_Methods_A_Review, https://journals.ekb.eg/article_276188_0.html, https://thearf-org-unified-admin.s3.amazonaws.com/MSI/2020/06/MSI_Report_16-106.pdf

take additional research for individual stores, and includes sellers at local flea markets as well. For sellers who are more hidden and do not have online information or reviews, they will be added whenever a buyer joins ThriftNet and adds stores after adding themselves as a buyer. Seller connections (aka edges between sellers) will then be calculated for this new node based on buyer descriptions and geographic proximity. Buyer nodes are connected through friends within the ThriftNet app, and if a buyer lists a friend, then an edge between the two buyers is created. Buyer-seller edges represent buyer engagement with a seller, such as visiting a store, writing a review, or making a purchase. Each buyer-seller edge also stores tags describing the interaction, including the reasons the user visited the seller or what they liked about the seller (e.g., specific styles, pricing, or unique items). This information will be gathered from buyer input whenever the buyer uses the app and signifies that they have a connection with a seller. Recommendations are generated by combining social exploration (which sellers a buyer's community engages with) with structural expansion across the seller network to identify related sellers. The prototype will include an interactive network visualization showing clusters of thrift sellers and how recommendations move through social and stylistic (in terms of fashion) neighborhoods. Also note that each node stores rudimentary information about itself, e.g., a seller node may include a few keywords such as "y2k," etc.

We will collect data about local thrift stores to construct a network graph as described above through the following:

1. Publicly available online listings and social media posts[11]
2. Store metadata such as location and aesthetic focus[12]
3. Online reviews[13] (such as in Figure 1)
4. Purchases made (specific to individuals as they use the app, we will curate these in a reasonable way for the prototype)

We will also analyze data for a specific buyer and their social circle, specifically their friends on the app we develop. For testing purposes, this will be simulated for different kinds of buyers, such as those with friends who are very in line with current fashion trends and popular "well-known" thrift stores, and those with friends who have more distinct fashion styles, uninfluenced by trends and social media.

Once the heterogeneous graph is constructed, we evaluate our network-aware solution for finding "hidden gem" thrift sellers against searching for items and trying to find "hidden gem" sellers online by systematically evaluating the ranking of the results to see whether the hidden gem store(s) even appear in the search results and how far down they are situated. In particular, we evaluate how structural properties of the network, such as centrality, clustering structure, and path length, affect the ability of buyers to reach previously undiscovered sellers. Recommendations begin with sellers that a buyer or their friends have already engaged with and expand outward through neighboring nodes in the seller network, allowing the system to suggest new sellers. Rather than relying on a single algorithm, we analyze several graph-based signals, including local network neighborhoods, communities identified through clustering, and influential sellers identified via centrality measures such as degree centrality and PageRank, which capture sellers that serve as important hubs or bridges for discovery within the thrift ecosystem. These will help us understand how social connections and structural relationships jointly shape network-informed discovery of new thrift stores. We will also study how recommendation performance varies across buyer profiles, including those who explicitly follow trends versus those with more individualistic styles (simulated for demonstration purposes). To illustrate the findings, the prototype will include a visualization that highlights various

[11] https://www.facebook.com/groups/WeirdSecondhandFinds, https://www.instagram.com/p/DSxsWYAD04A/
[12] https://www.thriftmaps.app/
[13] https://customerreviews.google.com/

clusters, social neighborhoods, and the various recommended exploration paths (graph traversal) for thrift stores, demonstrating how network structure influences the discovery of unique second-hand sellers beyond what keyword search alone provides.

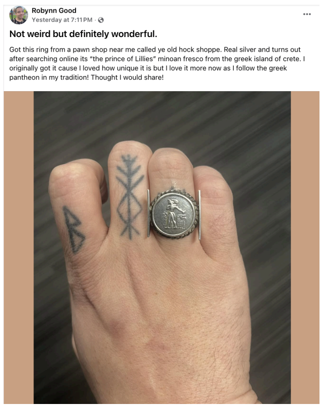


Figure 1: Example of a “hidden gem” find at a pawn shop called “ye old hock shoppe” that would require a lot of effort for someone to find with a keyword search alone.

End Product and Results

The expected outcome of this project is a prototype demonstrating how second-hand fashion discovery can be modeled and improved through network analysis. We aim to have developed a graph-based recommender system that represents thrift sellers and buyers within a heterogeneous network and evaluates how social and structural relationships influence discovery. The system will include an interactive visualization of the Pasadena/Los Angeles thrift network, highlighting different clusters as well as metrics that nodes in the cluster have in common, social neighborhoods, how recommendations may overlap amongst them, and how they are impacted, and recommended exploration paths to illustrate how buyers navigate the space.

Through controlled experiments, we will report comparisons between the traditional keyword-based search and network-aware recommendations that leverage both social connections and seller similarity. Our evaluation will examine how different types of buyers, including trend-focused buyers, buyers with distinct personal styles, and buyers unfamiliar with current trends, experience the discovery of second-hand sellers within the network. Rather than building a full marketplace platform, this project aims to provide a prototype showing that incorporating network structure can help buyers find “hidden gem” product sellers more efficiently and improve digital accessibility for thrift communities.

Proof of Concept

As a small proof of concept, we construct a toy network of buyers and sellers to illustrate how extracting network structure from real-world interactions can reveal meaningful discovery paths that are otherwise difficult to identify through traditional search. In the full application, each time a buyer joins the platform, they add themselves and the thrift stores they know to the graph, gradually expanding the network. For the proof of concept, we simulate this process by manually adding ourselves, several friends, and some additional imagined participants as buyer nodes (since our friendship circle is not very diverse), along with thrift stores that we are familiar with. Buyer-buyer edges represent friendship relationships, where two edges apart correspond to a “friend of a friend,” reflecting the social pathways through which thrift discoveries often spread. Seller nodes were constructed from thrift shops that we know of or have personally visited, and we supplemented these with publicly available information scraped from sources such as Yelp reviews, Instagram pages, and other online descriptions to capture store reputation, public perception, and geographic location. Edges between thrift shops are calculated using an algorithm that weighs geographic distance, inventory similarity, and buyer-reported satisfaction or reasons for visiting the store based on reviews and opinions. The edges between buyers and sellers indicate real-world interactions between a person and a store, such as visiting, browsing, or making a purchase. For this proof of concept, we create these edges using firsthand reports from our friends and ourselves, along with a small set of curated participants who intentionally visit specific stores. For each buyer-seller edge, we also incorporate information about why the buyer visited the store for readability purposes (this information would normally be input by the buyer when they use the app). Using this constructed network, we then demonstrate examples where a keyword search online does not reveal the object we want, but utilizing our network-based application allows us to discover the store through social connections and structural relationships between sellers.

Below are illustrations of our proof of concept network graphs.

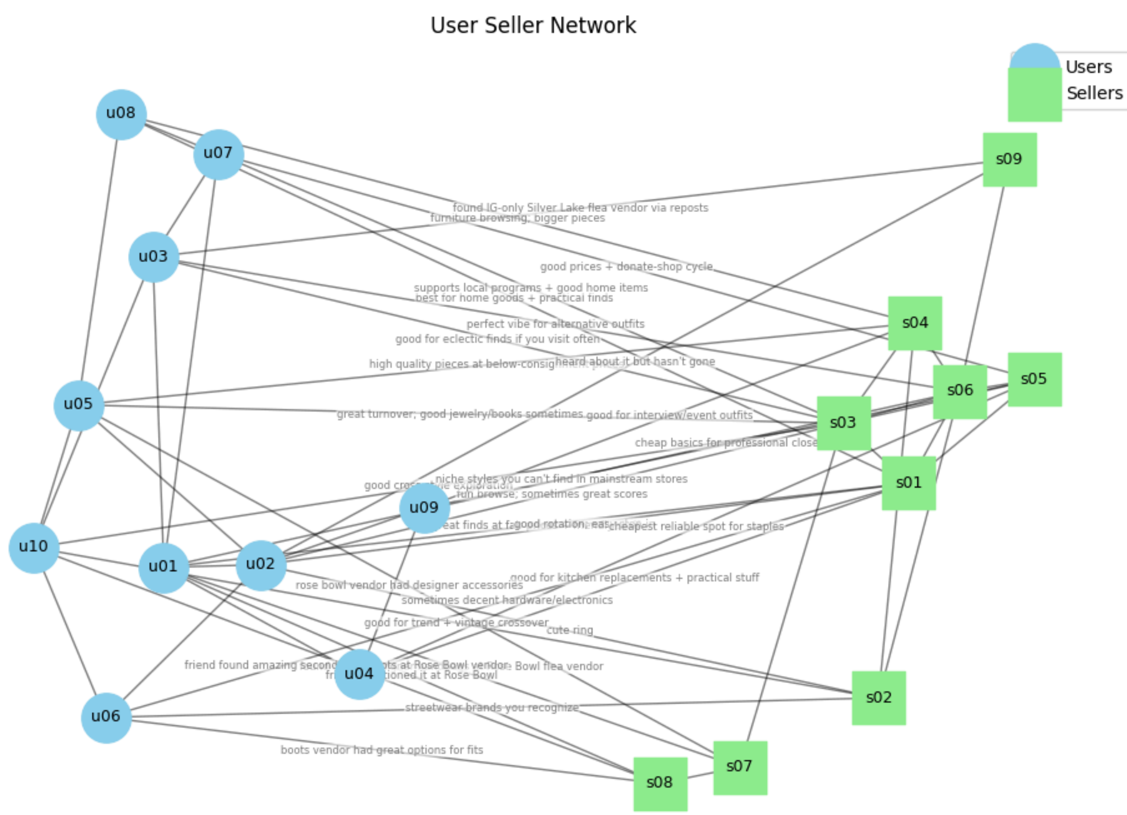


Figure 2: User-Seller Network. Illustrates the heterogeneous network constructed in our proof-of-concept system, consisting of two types of nodes: buyer nodes (blue) and seller nodes (green). The network integrates three layers of relationships: a buyer social network, a seller similarity network, and buyer-seller interaction edges. Buyer-buyer edges represent social relationships within our proof-of-concept user group and capture how fashion discoveries often spread through social ties. Buyer-seller edges represent real-world interactions between users and thrift sellers, including visiting, browsing, purchasing, or discovering a hidden gem. Each interaction stores contextual information describing why the buyer visited the store (for example, unique finds, vintage items, or good prices).

Seller-seller edges represent structural similarity between sellers and are created when stores share stylistic characteristics, geographic proximity, or similar seller types. This structure allows discovery to begin via graph traversal from sellers already visited by a buyer or their social neighbors and expand outward through related sellers in the network. In practice, discovery paths can be explored through graph traversal and ranking strategies such as neighborhood expansion (breadth-first search), shortest-path exploration, or centrality-based ranking signals.

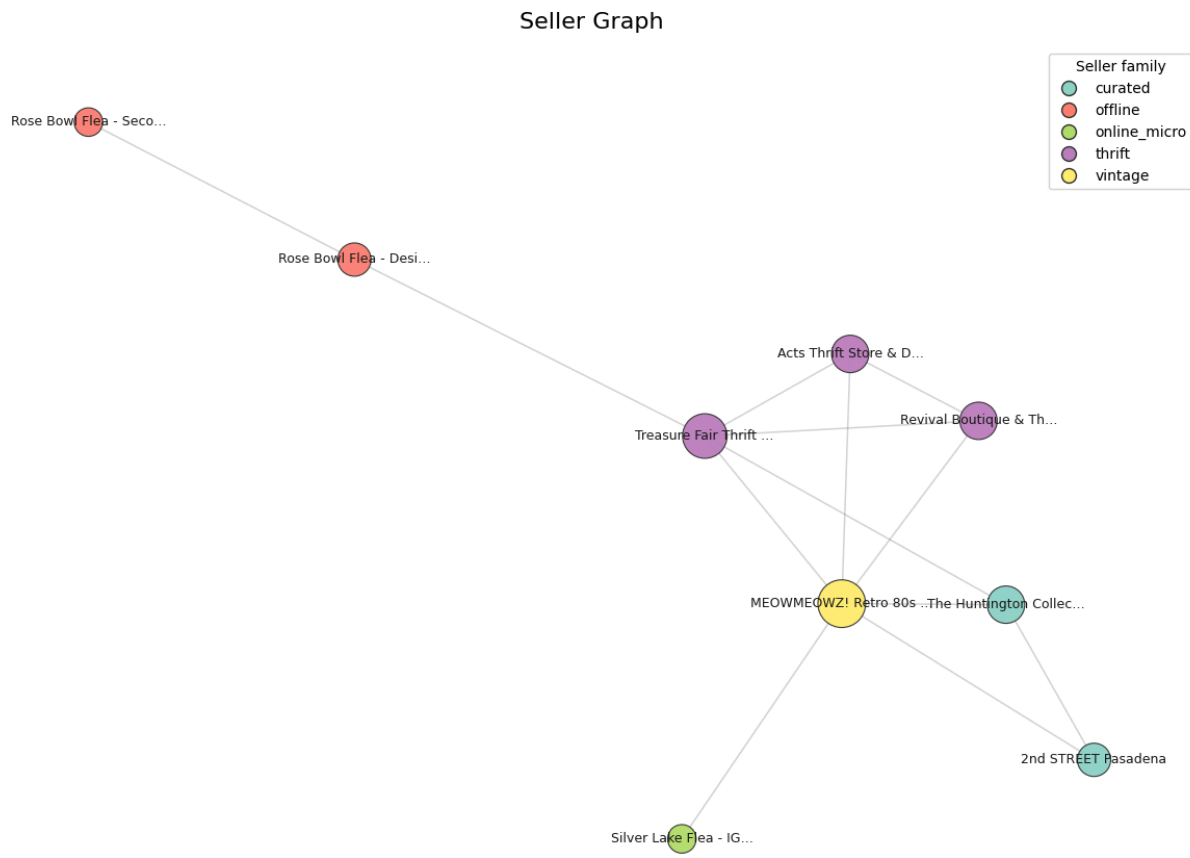


Figure 3: Seller-Seller Graph. Each node represents a seller and is color-coded according to a broad seller category (for example, thrift store, vintage boutique, flea market vendor, or online micro-seller). Edges represent similarity relationships between sellers and are created when stores satisfy one or more similarity conditions, such as sharing stylistic tags, operating within the same geographic area, or belonging to compatible seller categories. These connections approximate how buyers perceive similarity between stores when exploring the second-hand fashion ecosystem. The resulting network naturally forms clusters of sellers with similar styles or market positioning, which can be interpreted as communities within the resale ecosystem. These clusters support network-based exploration: once a buyer or their friend discovers a store within a cluster, the system can recommend nearby sellers in the similarity graph, enabling discovery of related stores that may not appear as directly in traditional keyword search results. To better capture real-world discovery patterns in the proof-of-concept setting, a small number of bridging connections are also introduced to represent socially discovered or offline sellers, such as flea market vendors that are often found through word-of-mouth.

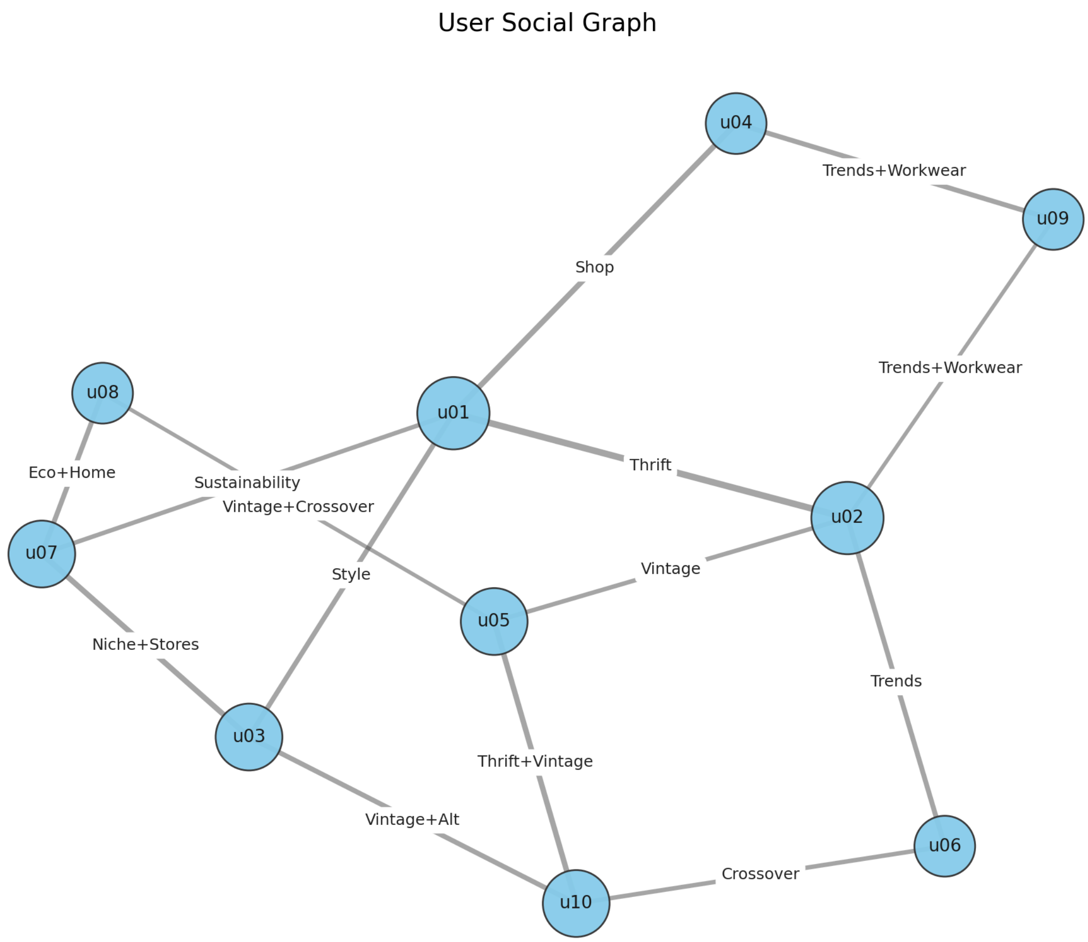


Figure 4: Buyer-Buyer Graph. Nodes represent buyers, and edges represent friendship or social relationships among participants in our dataset. Edge annotations summarize shared fashion interests or thrifting motivations between pairs of users (for example, “vintage,” “trends,” or “vintage + alternative”), providing contextual signals about why certain users influence one another’s store discovery behavior. This social layer models how second-hand fashion discoveries propagate through communities: buyers can explore stores visited by their direct friends as well as through multi-hop paths in the social network. By leveraging the small-world structure of social networks, where short paths exist between most pairs of nodes (“six degrees of separation”), the recommendation process is meant to mimic information diffusion in fashion communities, where trends and hidden finds often spread through word-of-mouth and social interaction.[14]

Below, we illustrate some examples of a hidden gem discovery using our proof-of-concept network.

Example 1

Sreeyutha is buyer 1, and Erica is buyer 2. Erica is looking for a very specific kind of unique black ring (a hidden gem) that she saw in a movie recently, but cannot find it online using a standard keyword search on the web. Because Erica and Sreeyutha are connected in the buyer social graph, Erica can explore stores that Sreeyutha has

[14] https://arxiv.org/abs/0706.0641, https://ijarsct.co.in/Paper19635proof-of-concept.pdf

visited recently. Erica observes that Sreeyutha visited a resale store known for trendy and unique accessories called 2nd STREET Pasadena, which Erica is not familiar with. Erica then goes to the store and finds an incredibly unique ring as below, which Erica would not have found by searching “unique black ring” online. Hence, Erica benefits from the friend connection with the ease of this discovery (optimal case). If Erica continues exploring the network, the system can further recommend structurally related sellers connected to that store in the seller similarity graph, enabling the discovery of additional shops that would not easily appear in standard search results.

Figure 5: The Unique Black Ring Erica found from 2nd STREET Pasadena.

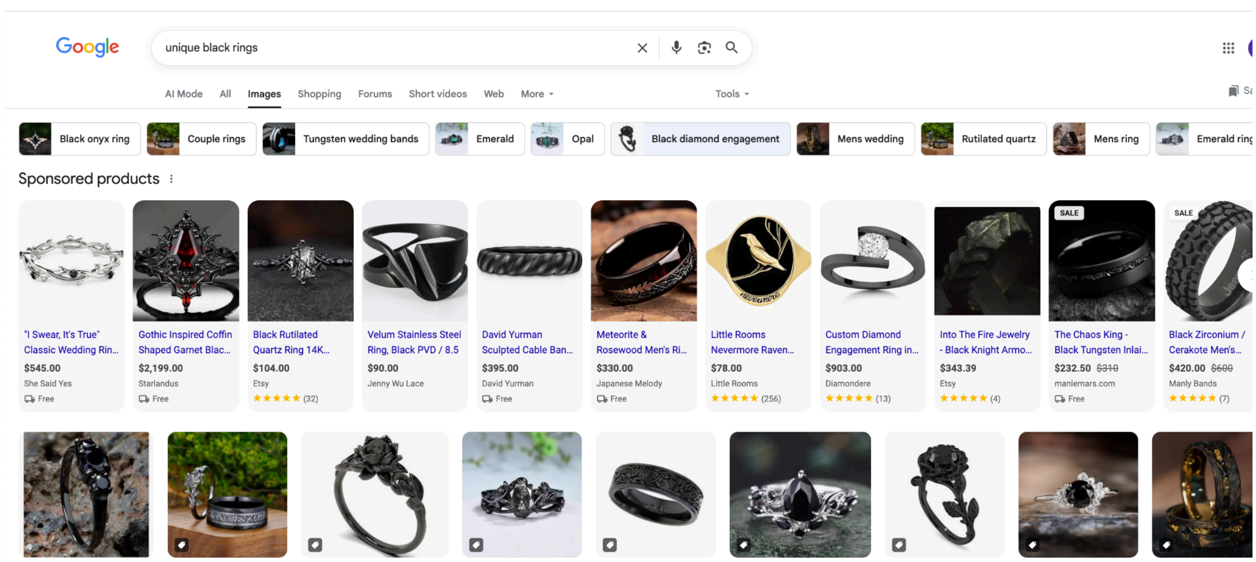


Figure 6: Erica’s attempt at finding the Unique Black Ring desired online with a keyword search, without taking into account social dynamics such as Erica’s friends.

Example 2: Punk overcoat via a friend-of-friend path (MeowMeowz / s06)

Suppose buyer 8 Maddy is searching for a punk-style overcoat (e.g., a black punk jacket at a great price), but can't easily find a great store for this on the web through keyword search. We see a lot of results from Depop, Etsy, Poshmark, and Ebay but these are online stores, and we miss local stores like Meowmeows, which would be a great find in this case. However, note that Meowmeowz does not have an online website and hence is generally hidden digitally. In our proof-of-concept buyer social graph, u08 is connected to u05, who is connected to u03, so u03 is a friend-of-a-friend of u08. Because our network treats two-hop connections as meaningful for discovery, u08 can explore stores that u03 has interacted with through this friend-of-friend path and discover MeowMeowz Retro 80s Thrift Boutique as a result. This reflects the small-world structure common in social networks, where short paths through friends-of-friends can rapidly expose users to new parts of the network.

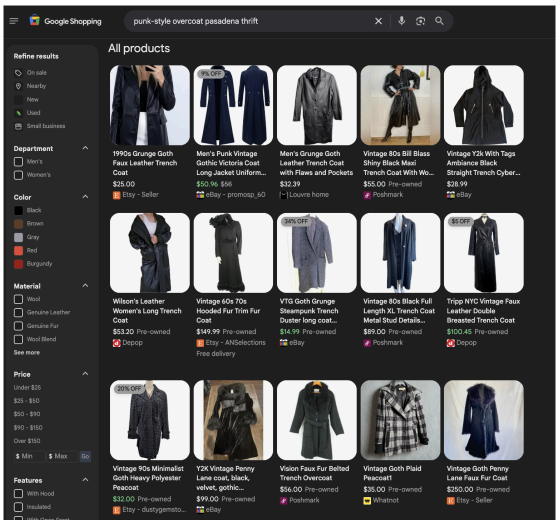


Figure 7: Maddy's attempt at finding a punk-style overcoat in Pasadena through thrifting. The style desired (shown in Figure 8) is difficult to find, as most stores we see are only online (Depop, Poshmark, eBay, etc).

Figure 8: The perfect black punk-style overcoat Maddy is looking for, that she found after visiting MeowMeowz in Pasadena in person!

Proof of concept code link: https://github.com/SreeyuR/cs144-proposal